MA JIANFA

(86) 133 2280 5128 | (852) 5735 4873 | mjf1287216989@gmail.com

Github: https://github.com/Pluviophil3

Research Interests: Embodied AI, Visual Language Navigation, Autonomous Driving, 3D Computer Vision

EDUCATION

Southern University of Science and Technology *Shenzhen, Sep 2022 – Jun 2026*

B.E. in Computer Science and Technology

- GPA: 3.6 / 4.0
- Core Courses: Artificial Intelligence, Intelligent Robotics, Operating Systems, Software Engineering, Computer Vision

The Hong Kong Polytechnic University *Hong Kong, Sep 2025 – Dec 2025*

Semester Exchange in Department of EEE

- Program: Artificial Intelligence Engineering
- Core Courses: Deep Learning, Multimedia Communication, Web System and Technology

National University of Singapore *Singapore, Jul 2024 – Aug 2024*

Summer Research Program in School of Computing

- Project: Focused on Visual Computing studies, with research on traffic sign recognition in extreme scenarios
- Grade: A-

PUBLICATIONS

G. Lan, **J. Ma**, Z. Zhao, Q. Qi, Y. Liu, Q. Hao. Gaussian4DOCC: A 3D Gaussian Occupancy Prediction with Temporal Information Extraction and Spatiotemporal Fusion. *IEEE Transactions on Automation Science and Engineering.* (Under Review, Sep. 2025) (Co-first Author)

X. Chen, **J. Ma**, B. Zhou. FIT: Fast Aerial Embodied Navigation via Structural-Semantic aware Planning. *IEEE Robotics and Automation Letters*. (Submitted, Sep. 2025)

RESEARCH EXPERIENCE

FIT: Fast Aerial Embodied Navigation via Structural-Semantic Aware Planning

Research Intern | Supervisor: Prof. Boyu Zhou *May 2025 – Present*

- OBJECTIVE – To develop a real-time aerial navigation system capable of objects navigation in large-scale, unknown outdoor environments based on natural language commands. The core goal was to bridge the gap between high-latency Vision-Language Models reasoning and the low-latency, continuous control required for aerial agent, enabling efficient zero-shot objects navigation
- METHOD – 1) *FIT Architecture*: designed a novel framework where the agent performs continuous, real-time replanning, with a VLM concurrently reasoning about high-level strategy asynchronously, which decouples slow VLM inference from fast UAV control; 2) *Semantic-aware Scene Representation*: implemented a module to select and retain keyframes with rich semantic value and creates a compact and informative scene memory for efficient VLM reasoning; 3) *Semantic-Geometric Consistent Planning*: developed a planning algorithm that integrates semantic guidance from VLMs with environmental geometric map, enabling continuous navigation
- ACHIEVEMENT – 1) Achieved a 78% navigation success rate in AirSim Simulation, representing a 204% performance improvement over the baseline model, and reaching 98% of human performance; 2) Completed navigation tasks with 37% of flight time compared to existing methods by using FIT framework to reduce reasoning latency and flight detours; 3) Validated effectiveness and feasibility of FIT on a custom-built quadrotor through over 10 hours of outdoor objects navigation experiments in diverse real-world scenarios
- CONTRIBUTION – 1) Conducted research on the latest work about Vision Language Navigation and proposed research question; 2) Designed and implemented a high-frequency geometric navigation path planner, a low-frequency VLM semantic scoring module and a ROS based cross-modules communication system; 3) Implemented baseline models and conducted ablation studies in AirSim simulation; 4) Designed and developed custom FIT quadrotor agent and conducted real-world experiment and demonstration

Gaussian4DOCC: A Spatiotemporal Fusion Gaussian Occupancy Prediction for Autonomous Driving

Team Leader | Supervisor: Prof. Qi Hao & Dr. Gongjin Lan *Sep 2024 – April 2025*

- OBJECTIVE – To develop a robust 3D Gaussian occupancy prediction model for vision-based autonomous driving in extreme environment by tackling the following limitations: 1) Inefficient Gaussian representation due to random initialization; 2) Imprecise prediction under occlusion caused by a single-frame framework; 3) Sparse supervision signals hindering early convergence
- METHOD – 1) *Historical Fusion*: utilized the prior historical Gaussian properties with temporal information to update the current Gaussians; 2) *Gaussian Features Aggregation*: proposed 4D Gaussian Sampling to incorporate both historical and current frames, thereby enhancing scene perception and reducing occlusion; 3) *Perspective Supervision*: leveraged auxiliary supervision signal to

improve accuracy and facilitate the convergence process of the training backbone

- ACHIEVEMENT – 1) Achieved a 32.20% IoU and 20.65% mIoU in occupancy prediction on nuScenes datasets, with a 7.95% and 8.12% relative improvement over baseline models; 2) Validated sub-modules effectiveness and contribution to overall performance by ablation study; 3) Validated robustness by quantitative and qualitative experiments on an extreme-task dataset.
- CONTRIBUTION – 1) Conducted related work research and identified limitations of existing work through experiments; 2) Proposed and implemented Gaussian4DOCC framework to tackle proposed limitations; 3) Performed model training, parameter tuning, and effectiveness evaluation through ablation studies and visualization analysis; 4) Led project management, including task allocation, milestone tracking, and reporting progress to supervisors; 5) Authored a research paper as a co-first author and submitted it to IEEE T-ASE (under review), with demo and codes released in Github;

Natural Language Instructions Based 3D Cross-Modal Perception System for UAVs

Core Member | Supervisor: Prof. Feng Zheng *Dec 2023 – Sep 2024*

- OBJECTIVE – To develop a generalized embodied UAV agent, leveraging VLM to enable natural language instruction and interaction
- METHOD – 1) *Perception & Mapping*: Processed RGB-D input with SLAM algorithm to generate 3D point clouds; 2) *Interaction*: Employed 3D-LLMs to process cross-modal inputs of environmental point clouds and user prompt, enabling user-agent interaction through question-answering about environmental information and object detection; 3) *Cognition & Decision-making*: Utilized tuned VLMs to generate waypoints from natural language inputs; 4) *Action*: Generated exploration path from discrete waypoints
- CONTRIBUTION – 1) Conducted research on 3DLLM, trained and finetuned models for UAV agent interaction tasks; 2) Finetuned VLM to generate waypoints for exploration tasks; 3) Developed the UAV platform's ROS system and deployed the model on UAVs

SCHOOL PROJECTS

Autonomous Limo Robot *Mar 2025 – May 2025*

SUSTech, Department of CSE, CS401 Intelligent Robot

- OBJECTIVE – To develop a Limo Robot with visual detection capabilities, and auto-exploration based on Visual Language Navigation
- TOOL – CV detection models, Visual Language Navigation, ROS, Nvidia Jetson AGX
- SOLUTION – 1) Deployed computer vision models to detect crosswalks, traffic lights, and road signs to aid in environmental decision-making; 2) Used Fast-SLAM for simultaneous mapping and localization during exploration; 3) Deployed a Visual Language Frontier Map model as the decision-making module for semantic navigation, which provided waypoints and feasible paths
- HIGHLIGHT – Our robot can efficiently search for unknown objects with deployed VLN models based on user's instruction.

Smart Album Software Development *Mar 2025 – May 2025*

SUSTech, Department of CSE, CS403 Software Engineering

- OBJECTIVE – To develop a web-based smart album application with features including image searching, clustering, and captioning
- TOOL – Flask, Spring Boot, Vue, SQL, object detection model, VLM, Git
- SOLUTION – 1) Built the backend using Spring Boot and the frontend with Vue and deployed models on a Flask server; 2) Implemented core functionalities such as object and face detection, CLIP-based image search and clustering, and VLM-powered Q&A and caption generation; 3) Utilized DevOps principles for collaboration, and automated testing and deployment using Git

COMPETITIONS & AWARDS

Pending Result, **Meituan 3rd Low-Altitude Economy Intelligent Flight Management Challenge** *Sep 2025*

Third Prize, **Guangdong Contemporary Undergraduate Mathematical Contest in Modelling 2023** *Dec 2023*

SKILLS

Languages – Fluent in English (TOEFL 102), Cantonese; Native in Mandarin

Programming – Python, Java, C/C++, MATLAB, LaTeX

Frameworks & Tools – ROS, MMCV, Unreal Engine, PyTorch, TensorFlow, Git, Nvidia Jetson

Areas of Expertise – Development of UAV Hardware-Software Platforms; Training and Deployment of Large Models; Design and Implementation of Navigation and Planning Algorithms; Processing and Operation of 3D Visual Data